

Research Task List: Monte Carlo Radiative Transfer and NICER Pulse-Profile Modeling

Undergraduate Research Project

Project Objective

The objective of this project is to construct a Monte Carlo radiative transfer model for neutron star surface emission, extract angular beaming functions, and incorporate these into synthetic NICER X-ray pulse-profile calculations. The ultimate goal is to quantify how radiative transfer-induced beaming alters pulse profiles and pulsed fractions relevant to NICER mass-radius inference.

1 Background Preparation and Literature Mastery

Tasks

- Study the physical principles of NICER pulse-profile modeling, including hot-spot geometry, relativistic light bending, and observer inclination.
- Understand how angular emission patterns (beaming functions) enter pulse-profile calculations.
- Identify where isotropic or parameterized beaming assumptions are used in current NICER analyses.

Required Reading

- Bogdanov et al. (2019), *ApJL*, 887, L25
<https://ui.adsabs.harvard.edu/abs/2019ApJL..887L..25B/abstract>
- Riley et al. (2019), *ApJL*, 887, L21
<https://ui.adsabs.harvard.edu/abs/2019ApJL..887L..21R/abstract>
- Miller et al. (2019), *ApJL*, 887, L24
<https://ui.adsabs.harvard.edu/abs/2019ApJL..887L..24M/abstract>

Concrete Outcome

A concise technical summary explaining how surface beaming affects NICER pulse amplitudes and inferred neutron star parameters.

2 Monte Carlo Radiative Transfer Implementation

Physical Model

- Plane-parallel neutron star atmosphere.
- Scattering-dominated opacity.
- Photon packets injected from the base of the atmosphere.

Algorithmic Tasks

- Initialize photon packets with isotropic upward directions.
- Sample optical depth to next interaction using

$$\Delta\tau = -\ln(\xi),$$

where $\xi \in (0, 1)$ is a uniform random variate.

- Convert optical depth to physical distance assuming constant opacity.
- At each interaction:
 - Draw a new propagation direction from the scattering phase function.
 - Optionally introduce anisotropy to mimic magnetic effects.
- Terminate photon packets when they escape the atmosphere.

Validation Requirements

- Verify photon number conservation.
- Demonstrate convergence of angular distributions with increasing packet count.

Required Reading

- Lucy (1999), *A&A*, 344, 282
<https://ui.adsabs.harvard.edu/abs/1999A&A...344..282L/abstract>
- Noebauer & Sim (2019), *Living Reviews in Computational Astrophysics*, 5, 1
<https://ui.adsabs.harvard.edu/abs/2019LRCA....5....1N/abstract>

3 Extraction of Angular Beaming Functions

Tasks

- Record the escape angle θ for each photon packet.
- Bin photons in $\mu = \cos \theta$.

- Construct the emergent intensity distribution

$$I(\mu) \propto \frac{N(\mu)}{\Delta\mu},$$

and normalize such that

$$\int_0^1 I(\mu) d\mu = 1.$$

- Repeat for multiple scattering prescriptions.

Required Reading

- Zavlin & Pavlov (2002), ASP Conf. Ser. 271
<https://ui.adsabs.harvard.edu/abs/2002ASPC..271..263Z/abstract>
- Ho & Lai (2001), *MNRAS*, 327, 1081
<https://ui.adsabs.harvard.edu/abs/2001MNRAS.327.1081H/abstract>

4 Construction of Synthetic NICER Pulse Profiles

Hot-Spot Geometry

- Assume one or two circular hot spots on the neutron star surface.
- Specify:
 - spot colatitude θ_s ,
 - observer inclination i ,
 - stellar compactness GM/Rc^2 .

Relativistic Light Bending

- For each rotational phase:
 - Compute the angle between the surface normal and the observer.
 - Apply Schwarzschild light bending to map emission angle to observer angle.
- Use standard approximations from NICER literature to relate emission angle α and impact parameter.

Incorporating Monte Carlo Beaming

- For each visible surface element:
 - Evaluate the Monte Carlo-derived beaming function $I(\mu)$ at the local emission angle.
 - Weight the observed flux contribution accordingly.
- Sum contributions over the hot spot to obtain observed flux as a function of rotational phase.

Pulse Profile Construction

- Compute the phase-resolved flux $F(\phi)$ over one rotation.
- Normalize to mean flux.
- Compute the pulsed fraction:

$$\text{PF} = \frac{F_{\text{max}} - F_{\text{min}}}{F_{\text{max}} + F_{\text{min}}}.$$

Comparison Models

- Repeat pulse-profile calculations assuming isotropic emission.
- Quantify differences in pulse shape and pulsed fraction.

5 Comparison with NICER Observations

Tasks

- Select well-studied NICER millisecond pulsars (e.g., PSR J0030+0451).
- Compare simulated pulsed fractions with published NICER values.
- Emphasize relative changes rather than full parameter fitting.

Scientific Goal

Demonstrate that radiative transfer-induced beaming can alter pulse-profile observables at levels comparable to NICER uncertainties.

6 Manuscript Preparation

Final Deliverables

- Monte Carlo beaming function plots.
- Synthetic NICER pulse profiles.
- Quantitative pulsed fraction comparison.
- A complete manuscript suitable for submission to *Research Notes of the AAS*.